Sanghyun Cho

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Education

Seoul National University — *Seoul, Republic of Korea*
Master of Science in Mechanical Engineering — Mar. 2025 – Present

- Advisor: Prof. Ho-Young Kim
- Thesis: *Pseudochaotic Tip Flutter of a Long Hanging Ribbon*

Seoul National University — *Seoul, Republic of Korea*
Bachelor of Science in Aerospace Engineering — Mar. 2018 – Feb. 2025

- Leave of Absence for Military Service (2 years)

Publications

Manuscripts

[M01] Sanghyun Cho, Jeongsu Lee, L. Mahadevan, and Ho-Young Kim. "Pseudochaotic Tip Flutter of a Long Hanging Ribbon." Under review (Physcal Review Letters **(PRL)**). 2026.

Conference

[P01] Sanghyun Cho, Jeongsu Lee, and Ho-Young Kim. "Pseudochaotic Tip Flutter of a Hanging Ribbon." *The 79th Annual Meeting of the APS Division of Fluid Dynamics* (**APS DFD**), Orlando, Florida, 2026. **(ORAL)**

[P02] Sanghyun Cho, Jeongsu Lee, and Ho-Young Kim. "Pseudochaotic Tip Flutter: Broadband Irregularity without Nonlinearity." *International Symposium on Interfacial and Multiphase Heat Transfer*, Seoul, Korea, 2026.

[P03] Sanghyun Cho, Jeongsu Lee, and Ho-Young Kim. "Pseudochaotic Tip Flutter of a Long Hanging Ribbon." *The 14th National Congress on Fluids Engineering*, Pyeongchang, Korea, 2026. **(ORAL)**

Research Experience

Microfluids and Soft Matter Laboratory, Seoul National University — *Seoul, Republic of Korea*
Graduate Researcher — Mar. 2025 – Present

Fluid–Structure Interaction and Nonlinear Dynamics

- Designed, fabricated, and assembled a custom vertical wind tunnel and modular test fixtures using CAD and 3D printing.
- Established repeatable ribbon-tracking conditions through pilot testing, integrating high-speed imaging, fluorescent coating, and controlled illumination.
- Conducted multiple wind-tunnel experiments to characterize different flutter dynamics.
- Developed MATLAB- and Python-based workflows for profile extraction, motion reconstruction, and data visualization.
- Derived physical scaling relations and developed a reduced-order aeroelastic model to explain the underlying dynamics, validating theoretical predictions against experimental data.
- Extended the fundamental research toward practical applications by identifying operating parameters and implementing a functional flutter-based ironing mechanism. **Bronze Prize [HA01]**
- Synthesized experimental and modeling results, prepared publication-quality figures, and drafted a first-author manuscript; presented findings through conference abstracts, an oral presentation, and a poster presentation. **[M01, P01–P03]**

Collective Robotics and Reconfigurable Systems

- Designing and fabricating parametric 3D-printed snake-like robots and implement decentralized, local-rule-based object grasping, enabling coordinated manipulation to emerge from simple module-level interactions.
- Developing geometric interlocking mechanisms for modular robots to investigate physical aggregation, reconfiguration, and the formation of mechanically connected three-dimensional structures.
- Building physics-based MuJoCo simulations to model inter-module contact and evaluate how local interaction rules give rise to collective and emergent behaviors.
- Developing a PyTorch-based simulation framework combining reinforcement learning (RL) and physics-informed neural networks (PINNs) to model how simple local interaction rules produce emergent collective behaviors for diverse manipulation tasks.

Honors & Awards

2025 **[HA01] Bronze Prize**, The 4th KSME–LG Electronics Future Home Tech. Challenge. Represented Team Ironing Man and developed "Flutter Ironing Technology," translating fluid–structure interaction research into a working ironing mechanism.

Project Experience

AI-Powered Home Beauty Device Development Spring 2026
Team Leader, Course Project (Capstone Design Project, Seoul National University) [Grade: A+]

- Led product planning and development of a functional prototype.
- Designed hardware using CAD and conducted repeated experiments to construct an operating regime map.
- Delivered the concept pitch and final presentation.

Flutter-Based Ironing Technology Development Jul. – Dec. 2025
Team Leader, Team Ironing Man, 4th KSME–LG Electronics Future Home Tech Challenge

- Managed six team members and coordinated development of a flutter-based ironing system.
- Designed and conducted experiments using high-speed imaging to characterize system performance.
- Delivered the final presentation; the team received the Bronze Prize.

Rapid-Application Tourniquet Development Spring 2025
Course Project (Introduction to Design of Medical Robots and Devices, Seoul National University) [Grade: A+]

- Developed a tourniquet incorporating a snap-through mechanism for rapid application in emergency settings.
- Designed experimental setups, conducted experiments, and collected performance data.

Smart Strength Training Machine Development Spring 2024
Team Leader, Course Project (Technology and Entrepreneurship, Seoul National University) [Grade: A+]

- Led project planning, prototype development, and application development.
- Proposed and refined the product concept and delivered the concept pitch and final presentation.

Mathematical Analysis on KZG Commitments and Proto-Danksharding Spring 2024
Course Project (Practical Applications of Blockchain 1, Seoul National University)

- Analyzed the mathematical foundations of KZG polynomial commitments and their role in Ethereum's EIP-4844 (Proto-Danksharding).
- Contributed mathematical analysis and explanations to a technical article produced by a four-member team. [Article]

Smart Plant Pot Development Sep. 2021 – Jun. 2022
Founder, CEO & CTO, Team ALOE

- Founded Team ALOE, recruited six team members, and led product planning, prototype development, and application development.
- Designed and developed an Arduino-based smart planter for nutrient, humidity, and temperature control, using CAD for hardware design and Python for programming.

Certifications

Aug. 2024 **Generative AI with Diffusion Models**, NVIDIA Deep Learning Institute

July 2024 **Deep Learning Basics**, NVIDIA Deep Learning Institute

July 2024 **Applications of AI for Anomaly Detection**, NVIDIA Deep Learning Institute

Dec. 2023 **Level 2 Sports Instructor, Bodybuilding**, Ministry of Culture, Sports and Tourism, Republic of Korea

Leadership & Service

Seoul National University Bodybuilding Team
Club President July 2024 – Feb. 2025
Executive Member July. 2021 – Feb. 2026

Seoul National University Department of Engineering Student Council
Student Council Member Mar. 2021 – Feb. 2023

Seoul National University Boxing Team *(Fist of SNU)*
Executive Member Mar. 2021 – June 2021

Aerospace Safety Agency, Republic of Korea Air Force
Battalion Representative July 2020 – Dec. 2020

Military Service

Republic of Korea Air Force *Republic of Korea*
Korean–English Translation Specialist, Sergeant May 2019 – Mar. 2021

- Performed Korean-to-English and English-to-Korean translation and interpretation during mandatory military service.
- Selected as Exemplary Soldier, recognized as a model service member within the Aerospace Safety Agency (2020).

Research Methods & Skills

Experimental Techniques	High-speed imaging, Experimental visualization & analysis, Motion tracking & analysis, Wind-tunnel testing
Programming & Tools	C/C++, Python, PyTorch, LaTeX
Simulation & Modelling	MATLAB, MuJoCo
Design & Prototyping	SolidWorks, 3D printing, Rapid prototyping, Experimental apparatus fabrication
Languages	Korean (Native), English (Full Professional Proficiency, TOEFL iBT: 5.5/6.0)